



SINOCHEN NINGBO LTD.

MATERIAL SAFETY DATA SHEET

SECTION 1. PRODUCT IDENTIFICATION:

Product name: Haloxyfop-P-methyl 108 g/L EC
Company Name: Sinochem Ningbo Ltd.
CAS RN.: 72619-32-0
Chemical name: Methyl (R)-2-[4-(3-chloro-5-trifluoromethyl-2-pyridyloxy) phenoxy] propanoate (IUPAC)
Formular: $C_{16}H_{13}ClF_3NO_4$
Product use: Herbicide
Formulation: Emulsifiable Concentrate

SECTION 2. COMPOSITION/INFORMATION ON INGREDIENTS

Material	Concentration
Haloxyfop-P-methyl	108 g/L min
Ingredients	To 1 L

Ingredients not precisely identified are proprietary or non-hazardous.

SECTION 3. HAZARDS IDENTIFICATION

Health Effects:

Acute:

Swallowed: The acute oral toxicity is low and small amount ingested incidentally during the handling of the concentrate and use of the prepared spray mix is not likely to be harmful.

Eyes: Solvent vapour may irritate the eyes. Concentrate in eye may cause moderate to severe irritation and possible corneal injury. The prepared spray mix may be slightly irritating to the eye.



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Skin Contact: The acute dermal toxicity is low and a single prolonged skin exposure is not likely to result in absorption of harmful amounts.

Inhalation: The acute inhalation toxicity is low and absorption of harmful amounts by inhalation is unlikely.

SECTION 4. EMERGENCY AND FIRST AID PROCEDURES:

Have the product container or label with you when calling a Poison Control Center or doctor, or going for treatment.

Inhalation:

If ill-effects occur move to fresh air. Contact a doctor.

Skin contact:

Remove contaminated clothing and wash skin thoroughly with soap and water. DO NOT scrub the skin.

Eye contact:

Immediately flush with cold water. Continue for at least 15 minutes. Contact a doctor.

Ingestion:

If swallowed, give water to dilute. DO NOT induce vomiting unless medical attention is delayed more than one hour. Take to a doctor or hospital immediately.

Notes to Physician:

Gallant NF contains petroleum-like solvents and the benefits of inducing vomiting must be weighed against the possibility of chemical pneumonitis. If lavage is performed, endotracheal and/or oesophageal control is advisable.

SECTION 5. FIRE AND EXPLOSION DATA:

Fire/Explosion Hazard

Flammable solvent in formulations. Combination of the vapour from the solvents with air may form an explosive mixture. Rapid decomposition of the active ingredient



occurs at temperatures greater than 300°C resulting in the formation of hazardous decomposition products. Drum explosion is likely if confined at high temperatures. Intact containers exposed to excessive heat should be cooled with water to reduce drum pressure. If possible use foam blanket to contain product. Avoid using AFFF. Contain runoff. Breathing apparatus is required.

SECTION 6. SPILL AND LEAK PROCEDURES:

Small Spill:

Wear appropriate protective clothing whilst cleaning up small spills (see Personal Protection). Apply absorbent material such as earth, sand or clay granules to the spill. Sweep up material for disposal when absorption is complete.

Large Spill:

Wear protective clothing and a respirator suitable for organic vapours. Clear the area of all unprotected personnel. As a standard precautionary measure, shut off all sources of ignition. Place leaking containers into salvage drums. Apply absorbent material such as earth, sand or clay granules to spill area. Form a barricade around the spill and in front of drains or waterways in spill vicinity, using earth or other available material. Prevent entry of material into drains or water ways.

SECTION 7. HANDING AND STORAGE DATA:

Storage and Transport

Not classified as a dangerous good. Store in tightly closed original containers in a locked, well ventilated place, out of direct sunlight. Avoid contact with food, clothing, drugs or other household goods during transport.

SECTION 8. EMPLOYEE PROTECTION RECOMMENDATIONS:

Engineering Controls

End User: No special controls required for normal conditions of use. Prepare spray in well-ventilated area.

Manufacturing: Atmospheric levels should be maintained below the recommended exposure guide-lines. Local exhaust ventilation may be necessary for some operations.



Personal Protection

End User: When preparing the spray, wear overalls buttoned to the neck and wrist, a washable hat and elbow length PVC gloves, face-shield or goggles. When using the prepared spray, wear overalls buttoned to the neck and wrist, a washable hat and face-shield or goggles. Wash hands after use. After each day's use wash gloves, face-shield or goggles and contaminated clothing.

Manufacturing: Wear overalls buttoned to the neck and wrist, a washable hat and elbow length PVC gloves, face-shield or goggles.

SECTION 9. PHYSICAL PROPERTIES:

Appearance:	Yellow brown to red brown liquid
Odor:	Light characteristic odor
Melting point:	Not available
Solubility in water:	Emulsifiable
S.g./density:	$1.010 \pm 0.005 \text{ g/cm}^3$
Vapor pressure:	Not available
Kow	technical: $\log P = 4.00$

SECTION 10. REACTIVITY DATA:

Stability: Stable under normal storage condition.

Hazardous Polymerization: Not pertinent.

Conditions to Avoid: Extreme temperature.

Hazardous Decomposition Products: Can decompose at high temperatures forming toxic gases.

SECTION 11. ANIMAL TOXICITY DATA:

Acute Toxicity Data for End Product:

Oral: LD50: rat > 2000 mg/kg



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Dermal:	LD50 (rat): > 2000 mg/kg
Inhalation:	LC50 (4 h) for rats > 4.74 mg/L air
Eye contact:	Moderately irritating to rabbit eye.
Skin Contact:	Slightly irritating to rabbit skin
Skin Sensitization:	Not a skin sensitizer

SECTION 12. ECOLOGICAL INFORMATION

For technical:

Birds: Acute oral LD50 for bobwhite quail 414 mg/kg.

Fish: LC50 (96 h) for rainbow trout 0.7 mg/l.

Algae: EC50 (96 h) >70 mg/l.

Bees: LD50 (48 h, oral and contact) > 100 µg/bee.

Worms: LC50 (14 d) 1343 mg/kg.

SECTION 13. DISPOSAL CONSIDERATIONS

Disposal:

Contaminated material must be disposed of in accordance with national, regional and district requirements.

Small Quantities and Containers:

Bury contaminated material at an approved landfill. Puncture top, bottom and sides of containers and bury below 500 mm in a disposal pit specifically marked and set up for this purpose, clear of waterways, vegetation and roots.

SECTION 14. TRANSPORT INFORMATION

UN No.:	3082
Hazard Class:	9
Packing group:	III
Proper shipping name:	Environmentally Hazardous Substance Liquid N.O.S.



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SECTION 15. REGULATORY INFORMATION

Designated name: Haloxyfop-P-methyl 108 g/L EC

Categories of danger: Harmful. Dangerous for the environment.

EC hazard: Xn - Harmful;

SECTION 16. OTHER INFORMATION

This MSDS summarises our best knowledge of the health and safety hazard information of the product and how to safely handle and use the product in the workplace. Each user should read this MSDS and consider the information in the context of how the product will be handled and used in the workplace including in conjunction with other products.